User Insights for Developing an Intelligent Course Scheduling System

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Experiment Design

The primary goal of the interview was to evaluate the current course scheduling systems, such as Workday Student, and uncover their limitations. Specifically, it aimed to determine user needs, frustrations, and whether an automated system for course scheduling would address these issues effectively.

The interviews were conducted in person to create a more relaxed environment, enabling participants to share their experiences openly and candidly. This setting also allowed for real-time follow-up questions to gain deeper insights.

Three voluntary participants were recruited, each providing informed consent. All participants, aged 22, are graduate students at Stevens Institute of Technology, representing diverse academic disciplines and sharing a South Asian background. Participant 1 is female and a Master's student in Information Systems, while Participant 2 and Participant 3 are male, pursuing Master's degrees in Engineering Management and Business Analytics, respectively. All participants are fluent in English, ensuring effective communication. The interviews lasted between 30 to 35 minutes, offering varied perspectives grounded in their unique educational experiences.

The questions were semi-structured, targeting:

1. Current course selection practices.
2. Satisfaction and frustrations with existing systems.
3. Thoughts on automated course scheduling.
4. Features necessary for trusting and using such a system.
   Questions were open-ended to encourage detailed responses while covering essential aspects of the user experience.

Experiment Results

Participants had varied course selection approaches, prioritizing timeslots, professors, and course content, with satisfaction ratings ranging from 2 to 8. Common frustrations included slow system response and lack of mobile compatibility. Scheduling time varied from 10 minutes to a week, with few finding the system satisfying. While none were aware of automated systems, opinions on their usefulness differed, emphasizing the need for transparency, preference input, and overrides. Participants suggested questionnaires and filters for sharing preferences. Course

withdrawal experiences ranged from easy but slow to tough, with one participant having no experience.

	Theme	P1	P2	P3
0	Course Selection Process	Time considerations	Not clear	Faculties and time slots, course preferences
1	Satisfaction with Current Systems	Low satisfaction (2/10), Slow system, Nothing satisfying	Moderate satisfaction (5/10), No app support, Nothing satisfying	High satisfaction (8/10), Loading process, Satisfied venues and schedules
2	Course Selection Priorities	Professors, Timeslots, Course content, Assignments	Professors, Grading generosity, Timeslots	Professors, Timeslots, Course content
3	Trust and Automation	Input detailed preferences, Suggestions only	Ability to override choices, Not helpful	Transparency, Input detailed preferences, Helpful
4	Experience with Automation	No experience with automation	No experience with automation	No experience with automation
5	Course Withdrawal Process	Easy but time-consuming, Transparency issues	Tough, Doesn't explain while dropping, Swapping easier	Never withdrawn

Overview of Participant Insights

The key insights reveal that dissatisfaction primarily arises from system slowness, a lack of mobile compatibility, and poor interface design. Preferences for automated systems differ, but most users favor options that are customizable and transparent. Additionally, the inclusion of features like filters and detailed questionnaires is crucial for gaining user acceptance of any automated system.

Your Reflection

Users prioritize customization and control over full automation, emphasizing the importance of transparency to foster trust. While inefficiencies exist in current systems, not all users see automation as the ideal solution, underscoring the need for balanced, user-centric designs.

The in-person format was valued for its ability to generate deep, free-flowing insights, though some participants' short responses highlighted the need to refine techniques for eliciting detailed feedback.